

GAOTONG®

GT30L24T3Y

Standard Chinese Font IC

— PRODUCT SPECIFICATION —

V1.0I_K
2024-10



www.hmi.gaotongfont.cn

Revision History

version number	Modify content	date	remarks
V1.0I_A	Customized chip specifications	2015-09	
V1.0I_B	Chip content modification	2015-09	
V 1.0I_C	Update font AC/DC parameters	2017-03	
V 1.0I_D	Power-up Timing 、 Delete sop8 series encapsulation	2019-06	
V1.0I_E	The clock frequency is updated from 50 MHZ to 45 MHZ	2019-07	
V1.0I_F	Remove the ASCII code 6X12	2019-08	
V 1.0I_G	Updated spec sheets	2020-03	
V1.0I_H	Update pin configuration and add four-wire instruction	2020-12	
V1.0I_I	Update the address and website 、 electrical character	2022-06	
V1.0I_J	Update the enterprise website website	2023-04	
V1.0I_K	Update 4.2、 4.3、 4.4	2024-10	

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1 General

GT30L24T3Y is a Chinese character library chip with 12x12 lattice, 16x16 lattice and 24x24 lattice. It supports Chinese characters of GB2312 national standard (including authorized by the National Beacon Commission), BIG5 character set and ASCII character. And compatible with Unicode encoding format. The arrangement format is vertical and horizontal. The user can directly read the dot matrix information of the inner code by using the function interface in the library file provided by our company.

1.1 Chip Features

- Data Bus: SPI serial bus interface
- Dot matrix arrangement: vertical byte, horizontal string
- Clock Frequency: 45MHz @ 3.3V
- Operating voltage: 2.7V ~ 3.6V
- Electric current:

Operating current: read current of 20mA (max)

Write-in current: 30mA (max)

Sleep current: 8uA (Max)

- Operating temperature: -40°C~85°C
- Package: DFN2x3/SOP8-B
- Character set: GB2312
BIG5
- 兼容 Unicode
- Font size: 12X12 16x16 24X24

1.2 Chip content

character set	Font library	word size	characters	form	Arrangement
ASCII character set	ASCII	5x7	96	standard	Y-vertical cross row
	ASCII	7x8	96	standard	Y-vertical cross row
	ASCII	8x16	96	standard	Y-vertical cross row
	ASCII	12 Unequal width	96	Round corner font	Y-vertical cross row
	ASCII	16 Unequal width	96	Round corner font	Y-vertical cross row
	ASCII	24 Unequal width	96	Round corner font	Y-vertical cross row
GB character set	GB2312 Chinese character	12x12	6763+470	Song typeface	Y-vertical cross row
		16x16	6763+470	Song typeface	Y-vertical cross row
		24x24	6763+470	Song typeface	Y-vertical cross row
	BIG5 Chinese character	12x12	5410+408	Song typeface	Y-vertical cross row
		16x16	5410+408	Song typeface	Y-vertical cross row
		24x24	5410+408	Song typeface	Y-vertical cross row
BIG5 Chinese internal code index table					
Unicode Chinese internal code index table					

1.3 Font Sample

1.3.1 Chinese Characters

Lattice	Form	standard	Font proofs
12	Song	GB	高通字库, 绽放文字之美
16	Song		高通字库, 绽放文字之美
24	Song		高通字库, 绽放文字之美
12	Song	BIG5	高通字庫, 綻放文字之美
16	Song		高通字庫, 綻放文字之美
24	Song		高通字庫, 綻放文字之美

1.3.2 ASCII dot matrix characters

Lattice	Form	standard	Font proofs
5X7	Standard	ASCII	AaBbCcDdEe12345
7X8	Bold		AaBbCcDdEe12345
12X12	Fillet		AaBbCcDdEe12345
8x16	Standard		AaBbCcDdEe12345
16X16	Fillet		AaBbCcDdEe12345
24X24	Fillet		AaBbCcDdEe12345

2 Operational order

2.1 Instruction Parameter

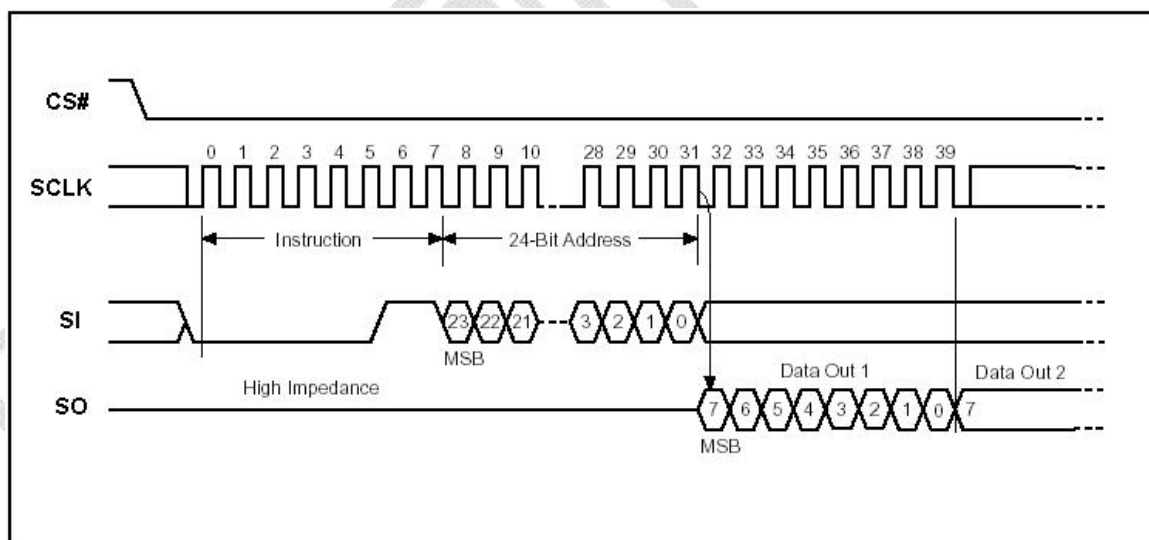
Instruction	Description	Instruction Code(One-Byte)		Address Bytes	Dummy Bytes	Data Bytes
READ	Read Data Bytes	0000 0011	03 h	3	—	1 to ∞
FAST_READ	Read Data Bytes at Higher Speed	0000 1011	0B h	3	1	1 to ∞

2.2 Read Data Bytes

The Read instruction supports up to 20 MHz, It outputs the data starting from the specified address location. The data output stream is continuous through all addresses until terminated by a low to high transition on CE#. The internal address pointer will automatically increment.

The Read instruction is initiated by executing an 8-bit command, 03H, followed by address bits [A23-A0]. CE# must remain active low for the duration of the Read cycle.

Figure: Read Data Bytes (READ) Instruction Sequence and Data-out sequence:

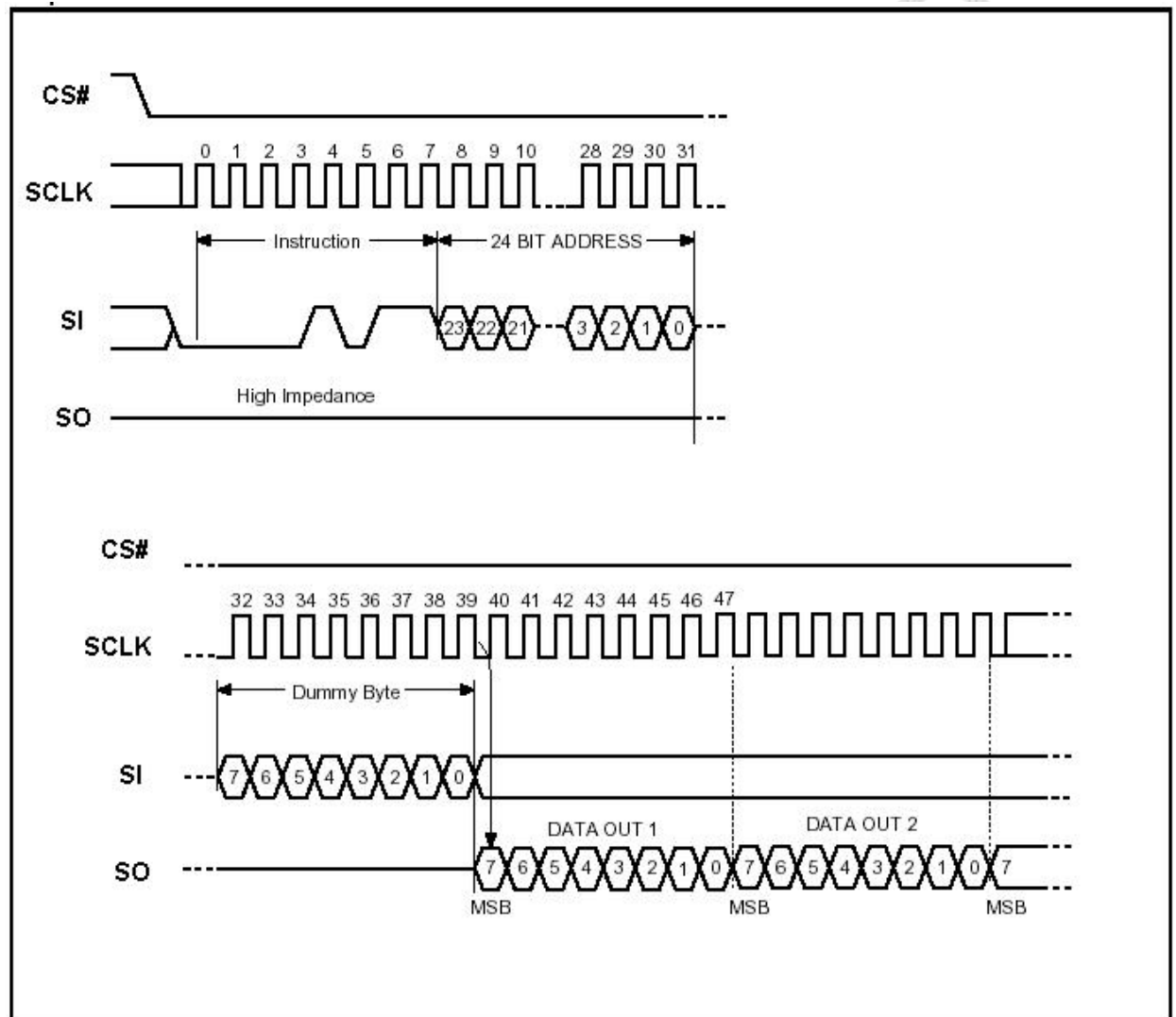


2.3 Read Data Bytes at Higher Speed

The High-Speed-Read instruction supporting up to 80 MHz is initiated by executing an 8-bit command, 0BH, followed by address bits [A23-A0] and a dummy byte. CE# must remain active low for the duration of the High-Speed-Read cycle.

Following a dummy byte (8 clocks input dummy cycle), the High-Speed-Read instruction outputs the data starting from the specified address location. The data output stream is continuous GT21L16T1W Standard Chinese Font Chip through all addresses until terminated by a low to high transition on CE#. The internal address pointer will automatically increment.

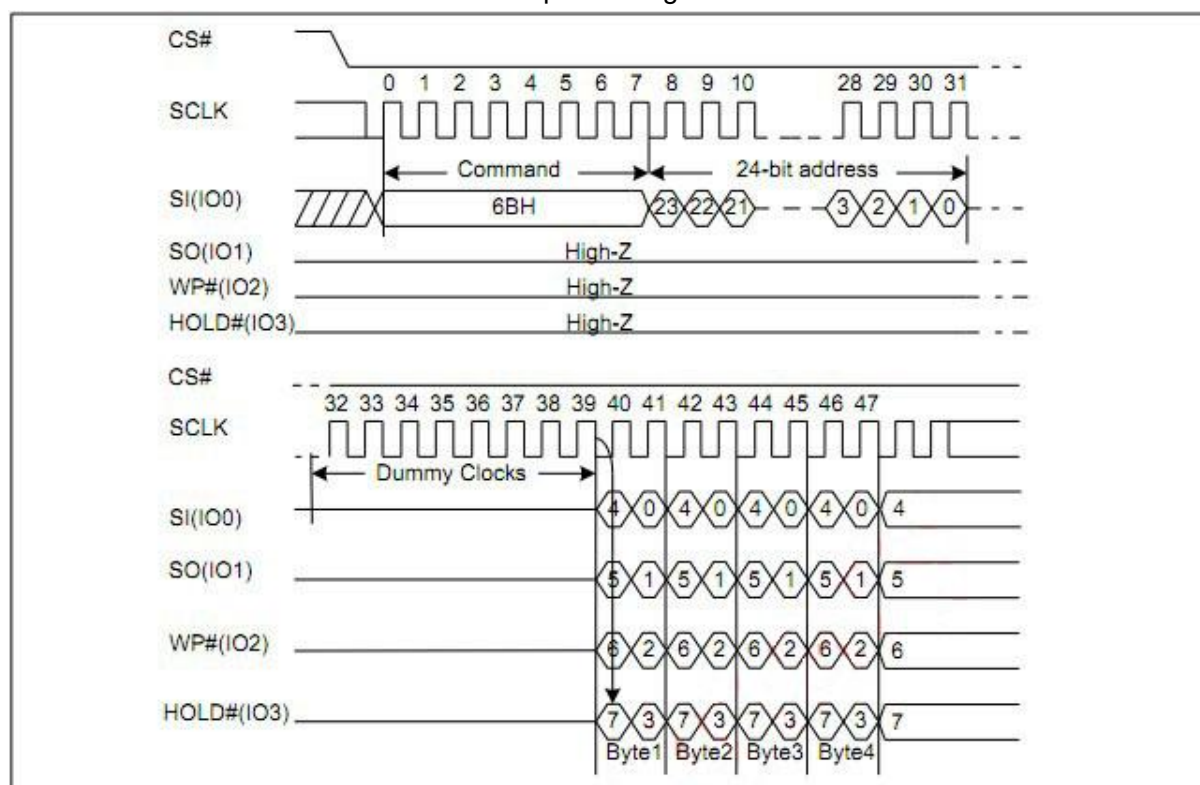
Read Data Bytes at Higher Speed (READ_FAST) Instruction Sequence and Data-out sequence:



2.4 Four-wire output fast read (6BH)

The four-wire output quick read command is followed by a 3-byte address (A23-A0) and an empty byte, each bit latched on the rising edge of the SCLK, and the memory content is then out of 4 bits from each clock cycle of the IO3,IO2,IO1,IO0. The command sequence is shown in the figure. the first byte addressed can be in any location. The address automatically increments to the next higher address after each byte of data is removed.

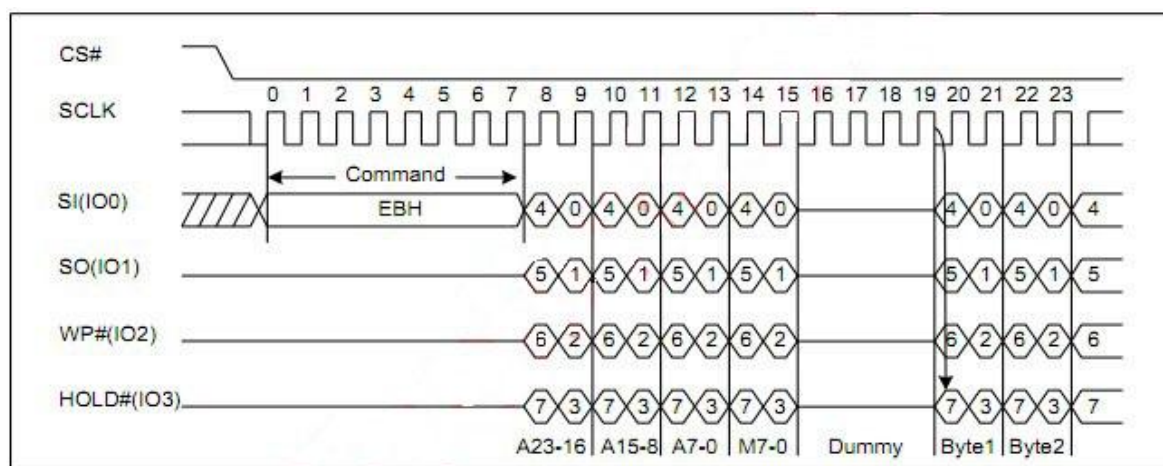
Four-wire output fast read
sequencediagram



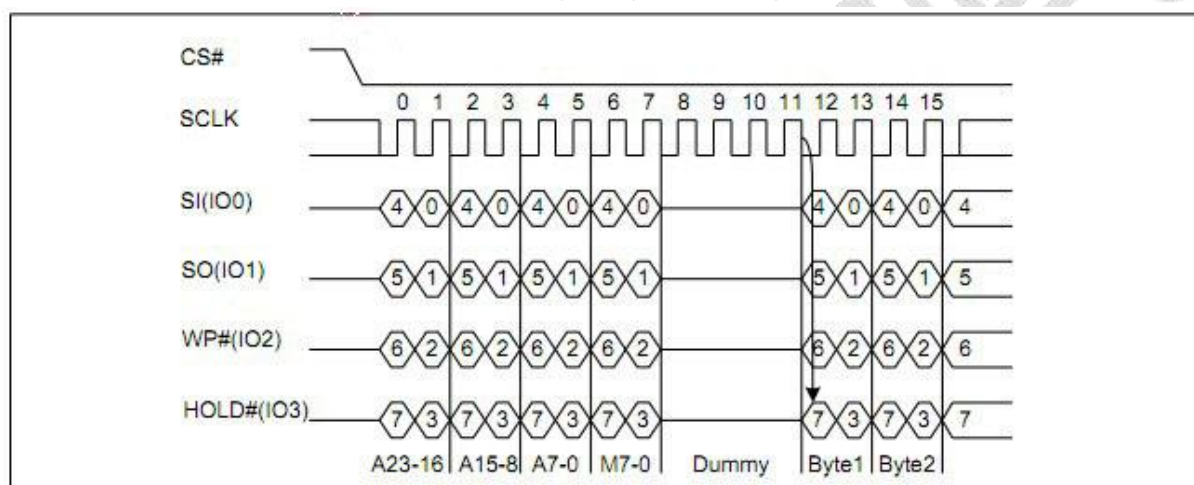
2.5 Fast reading I/O four-wire (EBH)

Four lines I/O quick read command, Enter 3 byte addresses (A23-0) and "continuous read mode" bytes and 4 empty cycle clocks, Each clock 4 bits each IO3, IO2, IO1, IO0 position is latched during the SCLK rise, IO3, the memory content IO2, IO1, IO0 4 bits per clock cycle, The command sequence is shown in the figure. the first byte addressed can be in any location. After each byte of data moves out, address automatically increments to the next higher address. the four-wire enable bit must set the status (QE) of the status register (S9) to enable.

Quad-line I/O fast read sequence diagrams (M7-0=0 XH or non- AXH)



Quick Read Sequence Diagram (M7-0=AXH) for Quad I/O (4)



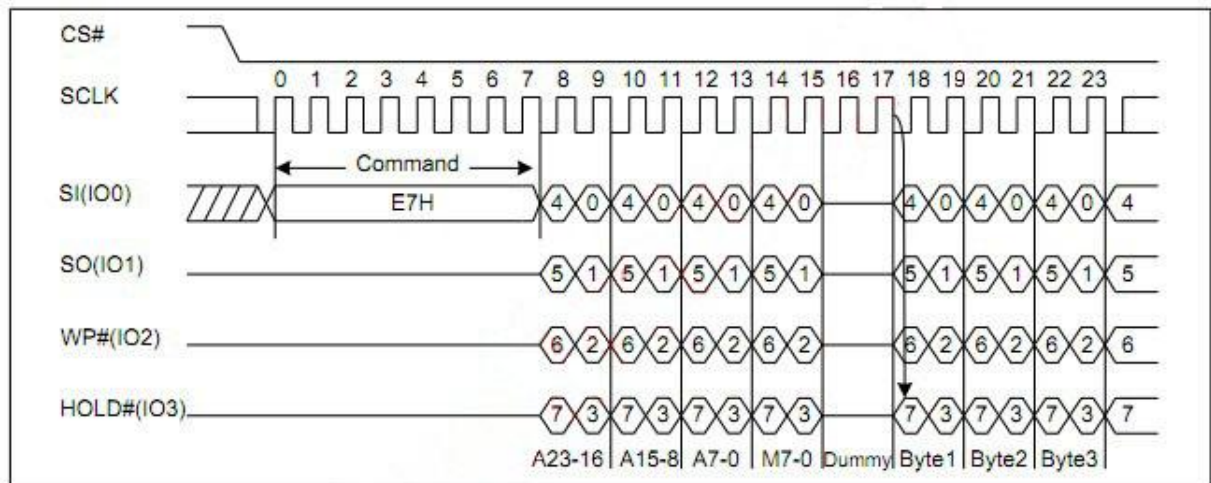
2.6 Fast reading of four-line I/O words (E7H)

The four-wire I/O word fast read command is similar to the four-wire I/O fast read command except that the lowest address bit (A0) must be equal to 0 and have only 2 pseudo clocks. The command sequence is shown in the figure. The first byte addressing can be in any location. The data that the address automatically increments to the next higher address byte is removed. the four enable bits (QE) of the status register (S9) must be set to enable four I/O words.

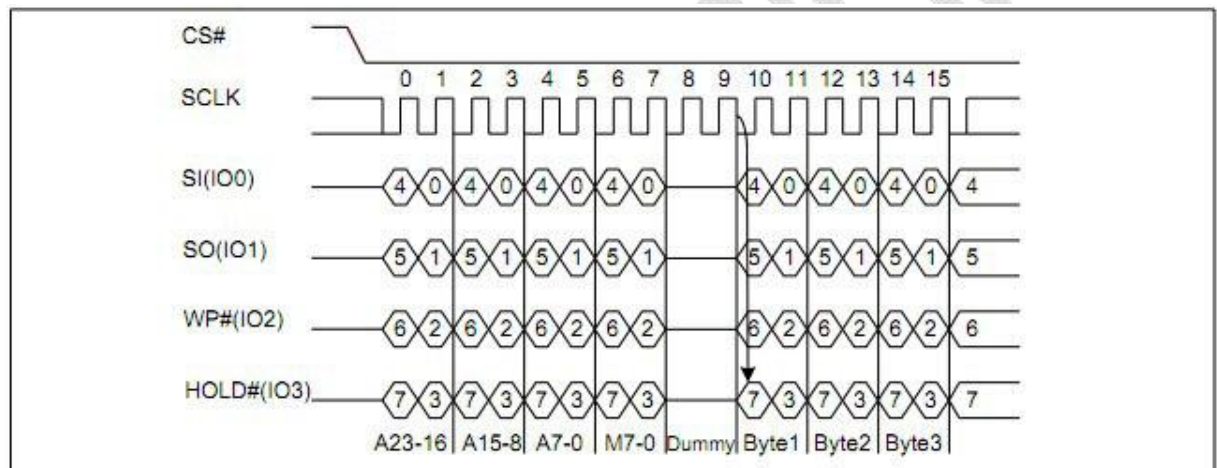
Four-wire I/O word fast read "continuous read mode"

By setting the "continuous read mode" bit (M7-0) after entering the 3 byte address (A23-A0), Fast reading commands I/O four lines can further reduce command overhead. When the Continuous Read Mode bit (M7-0)=AXH, The subsequent four-line I/O word fast read command (after the CS# is raised and lowered) does not require E7H command code. The command sequence is shown in the figure. if the "continuous read mode" bit (M7-0) is any value other than AXH, And the next command requires the first E7H command code, so return to normal operation. Continuous Read mode reset command can be used to reset before normal command is issued (M7-0).

Quick read sequence diagram (M7-0=0 XH or AXH)



Quick read sequence diagram (M7-0=AXH) for four-wire I/O word
(s)



2.7 Four-line page write instruction (32H)

A four-wire page write command is used to use four pins (IO0, IO1, IO2, and IO3) to program the memory. To use the four-line page write feature, the four-state enable in the status register Bit9 must be set (QE = 1). Before sending the page programming command, The write enable (WREN) command must be executed to set the write enable latch (WEL) bit. By driving CS# low level, And then the command code (32 H), three address bytes and at least one data byte on the IO pin, Enter a four-wire page write command.

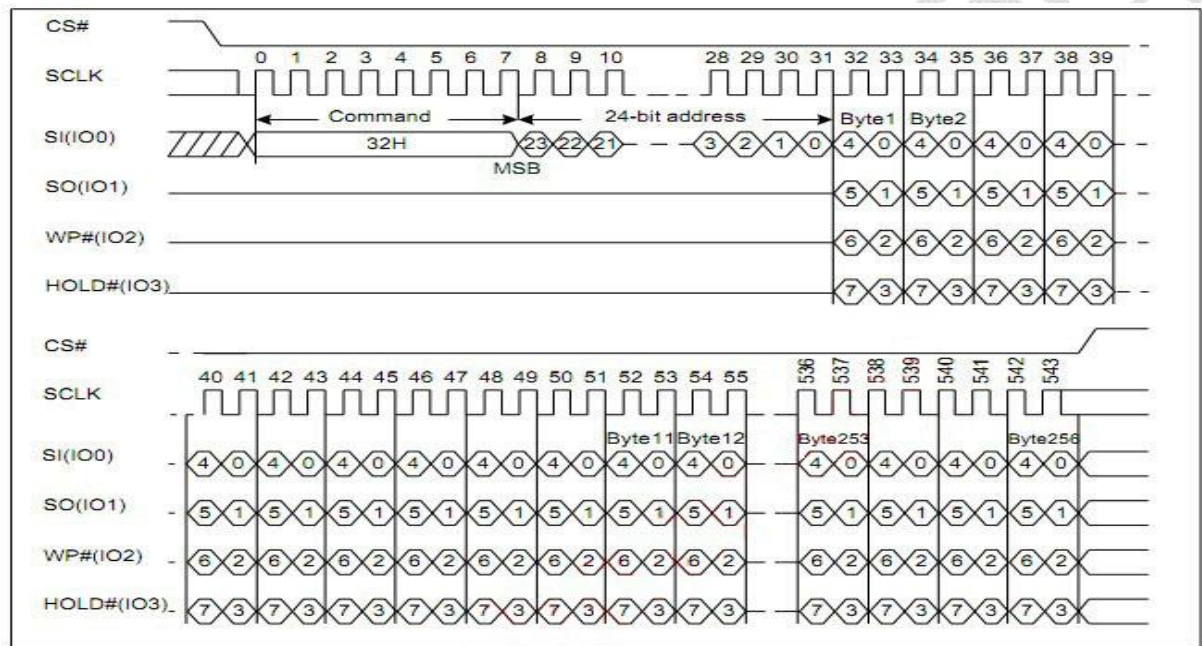
The command sequence is shown in the figure. If more than 256 bytes are sent to the device, the previously latched data is discarded, and the last 256 data bytes are guaranteed to be properly programmed within the same page. if less than 256 data bytes are sent to the device, they are programmed correctly at the requested address without any effect on other bytes on the same page. the CS# must be driven to a high level after the eighth bit of the last data byte is latched; otherwise, the four-wire page write (PP) command is not executed.

Once the CS# drive is high, the self-timing four-page programming cycle (its duration is t_{PP}) is started. a status register can be read to check the value of the write-in (WIP) bit when the

four-page programming cycle is in progress. During the self-timing four-page programming cycle, the write (WIP) bit is 1 and 0 after completion. At an unspecified time before cycle completion, write enable latch (WEL) bit reset.

One does not execute a four-page programming command applied to pages protected (BP4,BP3,BP2,BP1 and BP0) by blocks.

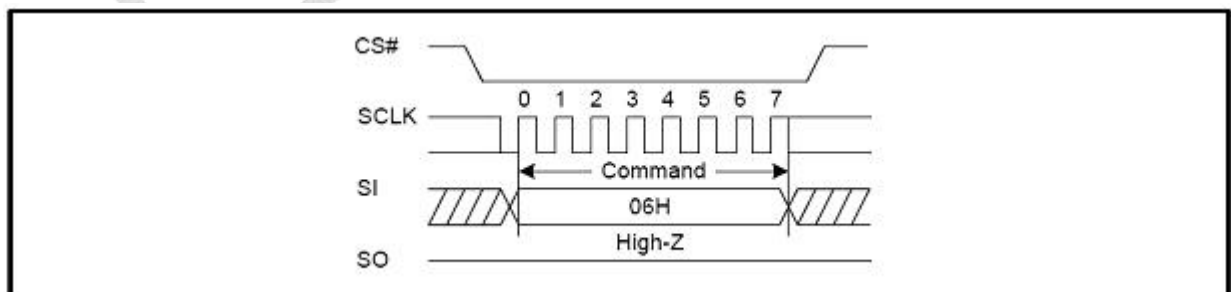
Quad page program sequence
diagram



2.8 Write Enable

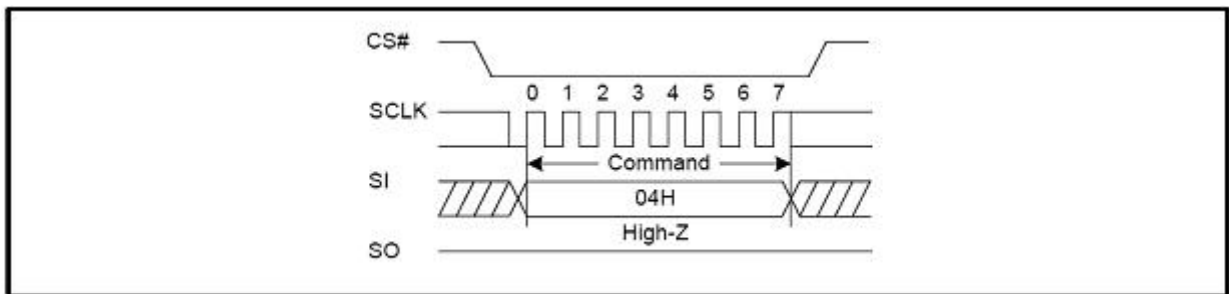
Write Enable sequence of instructions is as follows (figure):

CS# lower -(sends Write Enable commands ->CS# higher



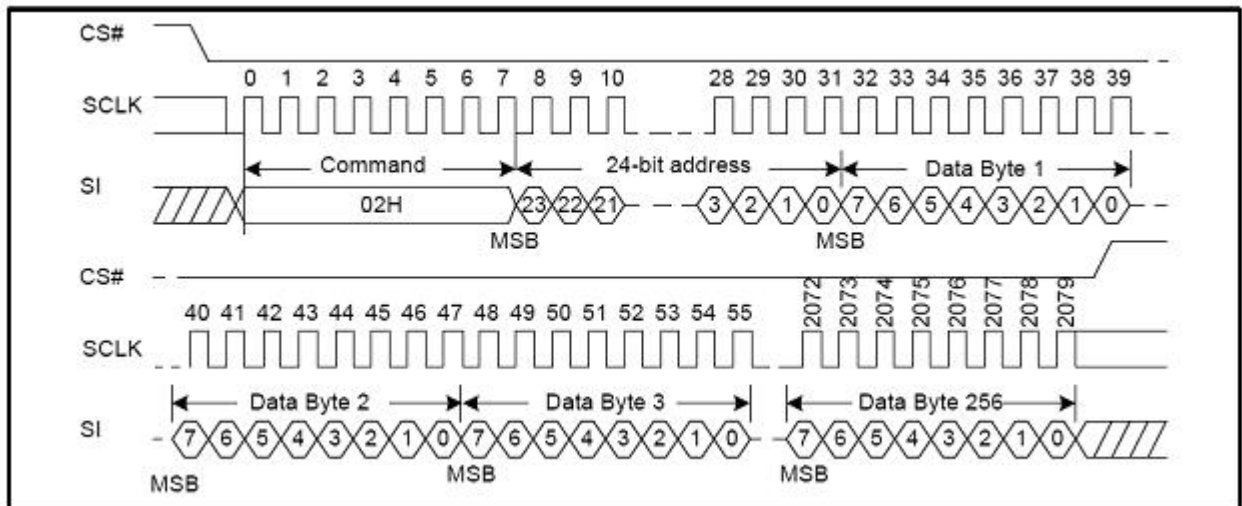
2.9 Write Disable

Write Enable sequence of instructions is as follows (figure):
CS# lower -(sends Write Disable commands ->CS# higher



2.10 Page Program

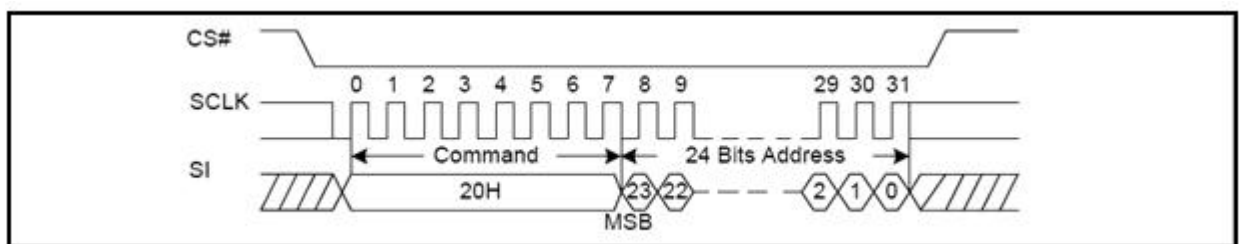
Page Program sequence of instructions is as follows (figure):
Program command (send 3 byte address -> send data ->CS# higher



Note: after the writing instruction is sent CS# higher, it is necessary to judge the busy state. After waiting for the chip to complete the writing, the next operation can be carried out on the chip. If there is no library file, please ask for it.

2.11 Sector Erase

Sector Erase sequence of instructions is as follows (figure):
CS# lower -(sends Sector Erase command (sends 3 byte address ->CS# higher

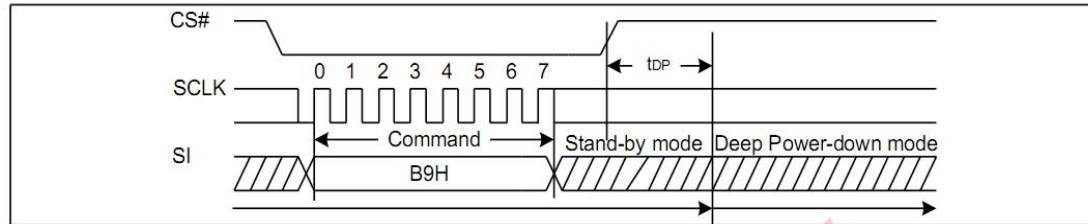


Note: after the erasure instruction is sent CS# higher, it is necessary to judge the busy state, wait for the chip to complete the erasure, before the next operation of the chip can be carried out. If there is no library document, please ask for it.

2.12 Deep Sleep mode command (B9H)

Once the font chip into a deep sleep mode, all commands are ignored, except wake deep sleep mode instruction, first CS # low, input B9H command, and then CS # goes high and the time duration TDP (TDP = 25us), within the duration of TDP, font chip enters deep Power Down mode.

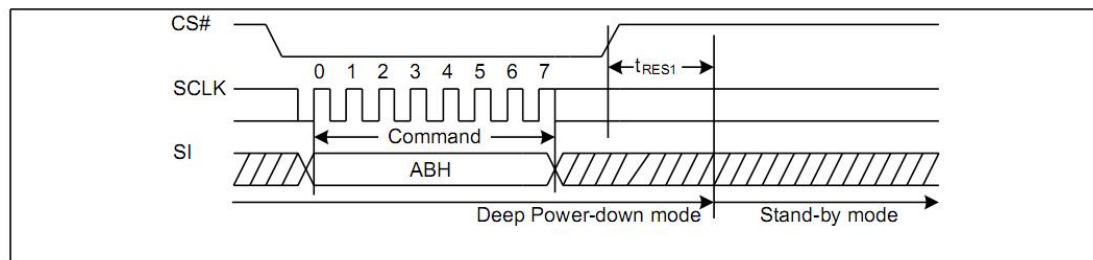
Deep Sleep mode commands Timing aveforms



2.13 Wake deep sleep mode instruction (ABH)

First CS # low, sends commands to the font ABH chip, then CS # goes high and continues Tres1 time ($T_{res1} = 25\mu s$), font chip will resume normal operation, CS # pin must be within Tres1 time remains high.

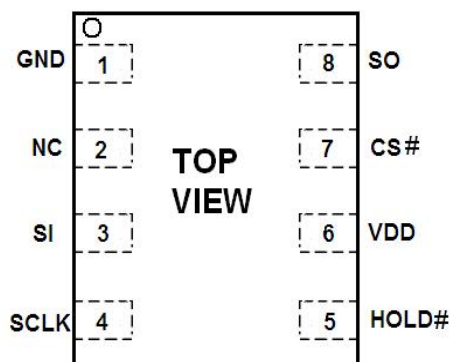
Wake deep sleep mode instruction timing waveform diagram



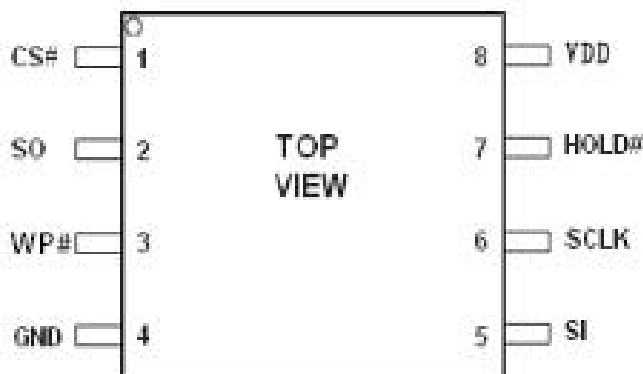
3 Pin Description And Circuit Connection

3.1 Pin Configuration

DFN2X3



SOP8-B



3.2 Pin Description

DFN2X3

NO.	name	I/O	Description
1	GND		Ground
2	NC		NC
3	SI	I	Serial data input
4	SCLK	I	Serial clock input
5	HOLD#	I	Hold, to pause the device without
6	VDD		+ 3.3V Power Supply
7	CS#	I	Chip enable input
8	SO	O	Serial data output

SOP8-B

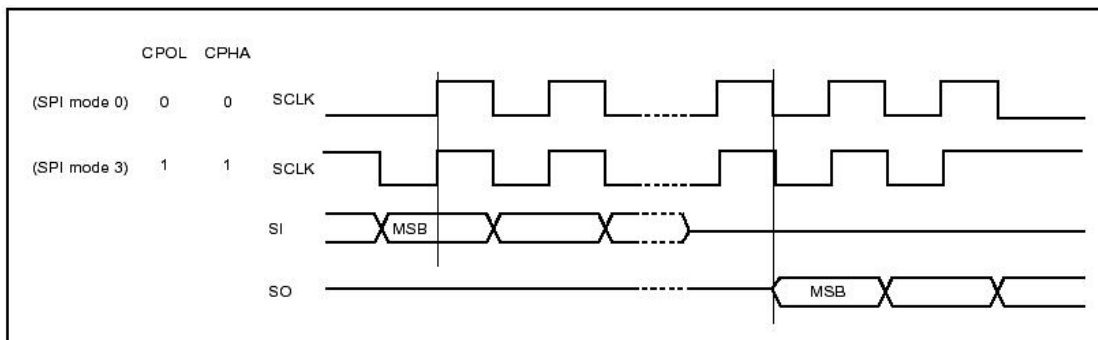
NO.	name	I/O	Description
1	CS#	I	Chip enable input
2	SO	O	1.Serial data output 2.Data input output1
3	WP#		1. Serial data output 2. Data input output2
4	GND		Ground
5	SI	I	1.Serial data input 2.Data input output0
6	SCLK	I	Serial clock input
7	HOLD#	I	1.Hold, to pause the device without 2.Data input output3
8	VDD		+ 3.3V Power Supply

Serial data output(SO): Data is shifted out on the falling edge of the serial clock.

Serial data input(SI) : Inputs are latched on the rising edge of the serial clock.

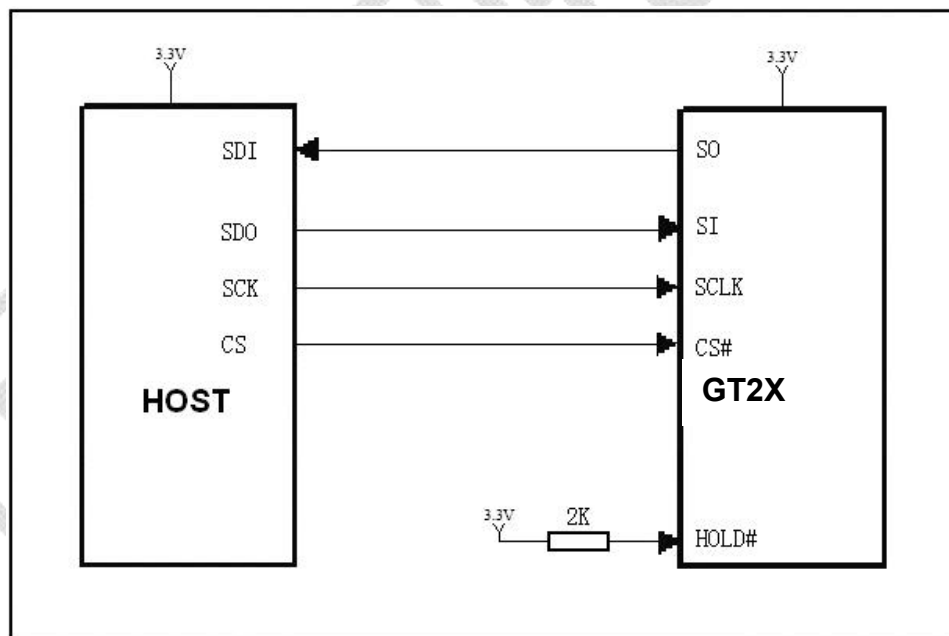
Serial clock input(SCLK): To provide the timing of the serial interface.

Chip enable input(CS#): The device is enabled by a high to low transition on CS#. CS# must remain low for the duration of any command sequence.



3.3 SPI Connection Block Diagram

SPI host interface circuit connection can refer to the following figure (#HOLD pin pulled recommend pick 2K resistor 3.3V).



SPI interface with host interface reference circuit schematic

4 Electrical Characteristics

4.1 Absolute maximum rating

Symbol	Parameter	Min.	Max.	Unit	Condition
T _{OP}	Operating Temperature	-40	85	°C	
T _{STG}	Storage Temperature	-65	150	°C	
V _{DD}	Supply Voltage	-0.3	3.6	V	
V _{IN}	Input Voltage	-0.3	V _{DD} +0.3	V	

4.2 DC

Condition: T_{OP} = -40°C to 85°C, GND=0V in SPI mode;

Symbol	Parameter	Min.	Typ.	Max.	Unit	Condition
I _{DD}	VDD Supply Current(active)	-	-	30	mA	
I _{SB}	VDD Standby Current	-	-	55	uA	/CS=VDD, VIN=VDD or VSS
I _{CC2}	Deep Power-Down Current	-	-	8	uA	/CS=VDD, VIN=VDD or VSS
V _{IL}	Input LOW Voltage	-0.5	-	0.16VCC	V	VDD=2.7~3.6 V
V _{IH}	Input HIGH Voltage	0.8VCC	-	VCC+0.4	V	
V _{OL}	Output LOW Voltage		-	0.4 (I _{OL} =1.6mA)	V	
V _{OH}	Output HIGH Voltage	VDD-0.2 (I _{OH} =-100uA)	-		V	
I _{LI}	Input Leakage Current	-	-	±2	uA	
I _{LO}	Output Leakage Current	-	-	±2	uA	

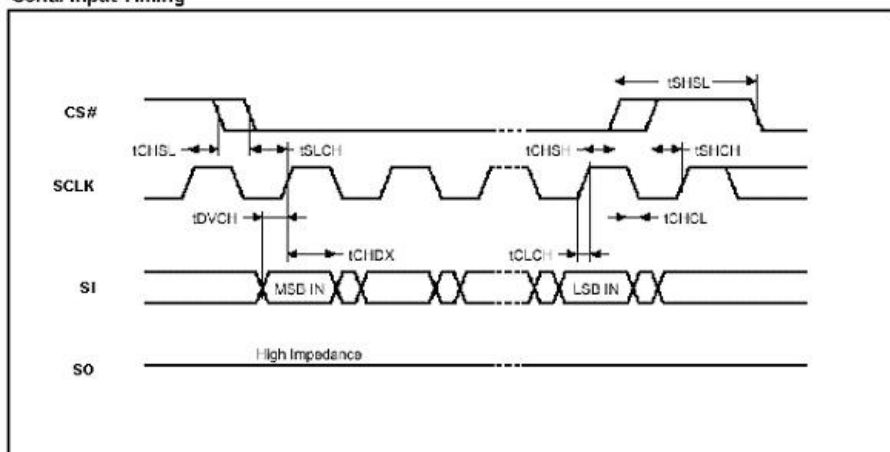
Note: I_{IL}: Input LOW Current, I_{IH}: Input HIGH Current,
I_{OL}: Output LOW Current, I_{OH}: Output HIGH Current,

4.3 AC

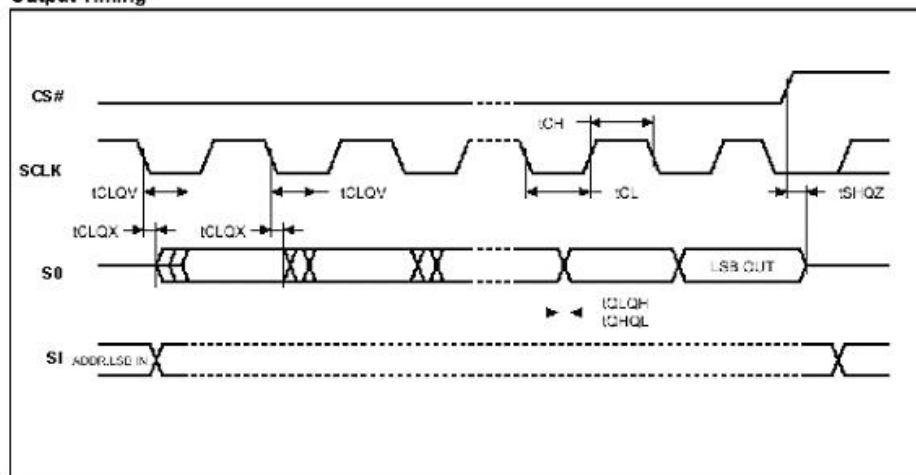
Symbol	Alt.	Parameter	Min.	Typ.	Max.	Unit
F _c	F _c	Clock Frequency	D.C.		45	MHz
t _{CH}	t _{CLH}	Clock High Time	5			ns
t _{CL}	t _{CLL}	Clock Low Time	5			ns
t _{CLCH}		Clock Rise Time(peak to peak)	0.2			V/ns
t _{CHCL}		Clock Fall Time (peak to peak)	0.2			V/ns
t _{SLCH}	t _{CSS}	CS# Active Setup Time (relative to SCLK)	7			ns
t _{CHSL}		CS# Not Active Hold Time (relative to SCLK)	5			ns
t _{DVCH}	t _{DSU}	Data In Setup Time	2			ns
t _{CHDX}	t _{DH}	Data In Hold Time	5			ns
t _{CHSH}		CS# Active Hold Time (relative to SCLK)	5			ns

tSHCH		CS# Not Active Setup Time (relative to SCLK)	5			ns
tSHSL	tCSH	CS# Deselect Time	130			ns
tSHQZ	tDIS	Output Disable Time			9	ns
tCLQV	tV	Clock Low to Output Valid			7.5	ns
tCLQX	tHO	Output Hold Time	5			ns
tHLCH		HOLD# Setup Time (relative to SCLK)	5			ns
tCHHH		HOLD# Hold Time (relative to SCLK)	5			ns
tHHCH		HOLD Setup Time (relative to SCLK)	5			ns
tCHHL		HOLD Hold Time (relative to SCLK)	5			ns
tHHQX	tLZ	HOLD to Output Low-Z			7	ns
tHLQZ	tHZ	HOLD# to Output High-Z			12	ns

Serial Input Timing

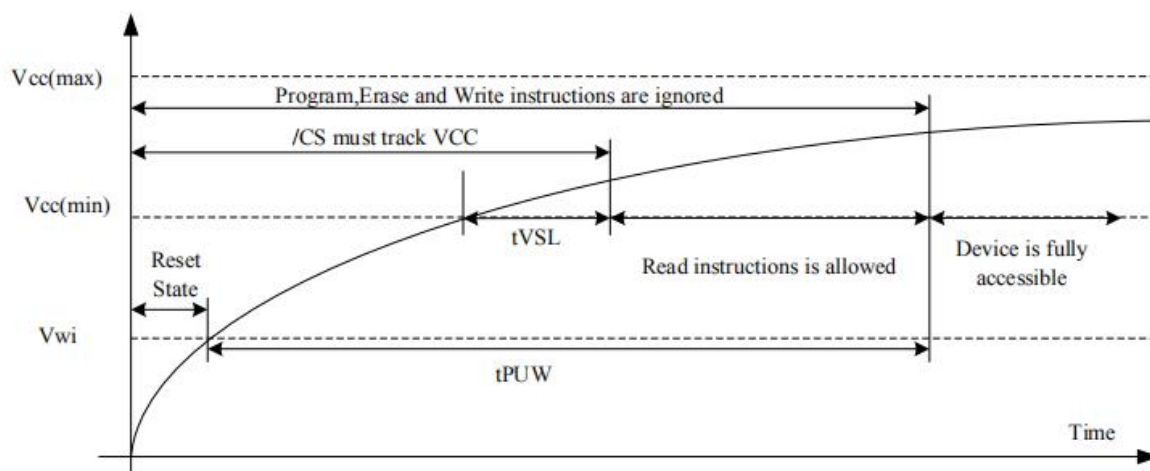


Output Timing



4.4 Power UP Timing

Symbol	Parameter	Min	Max	unit
T VSL	VCC(min)To/CS Low	300	-	us
T PUW	Time Delay From VCC(min)To Write Instruction	4	10	ms
VWI	Trite Inhibit Voltage VCC(min)	1.5	2.5	v

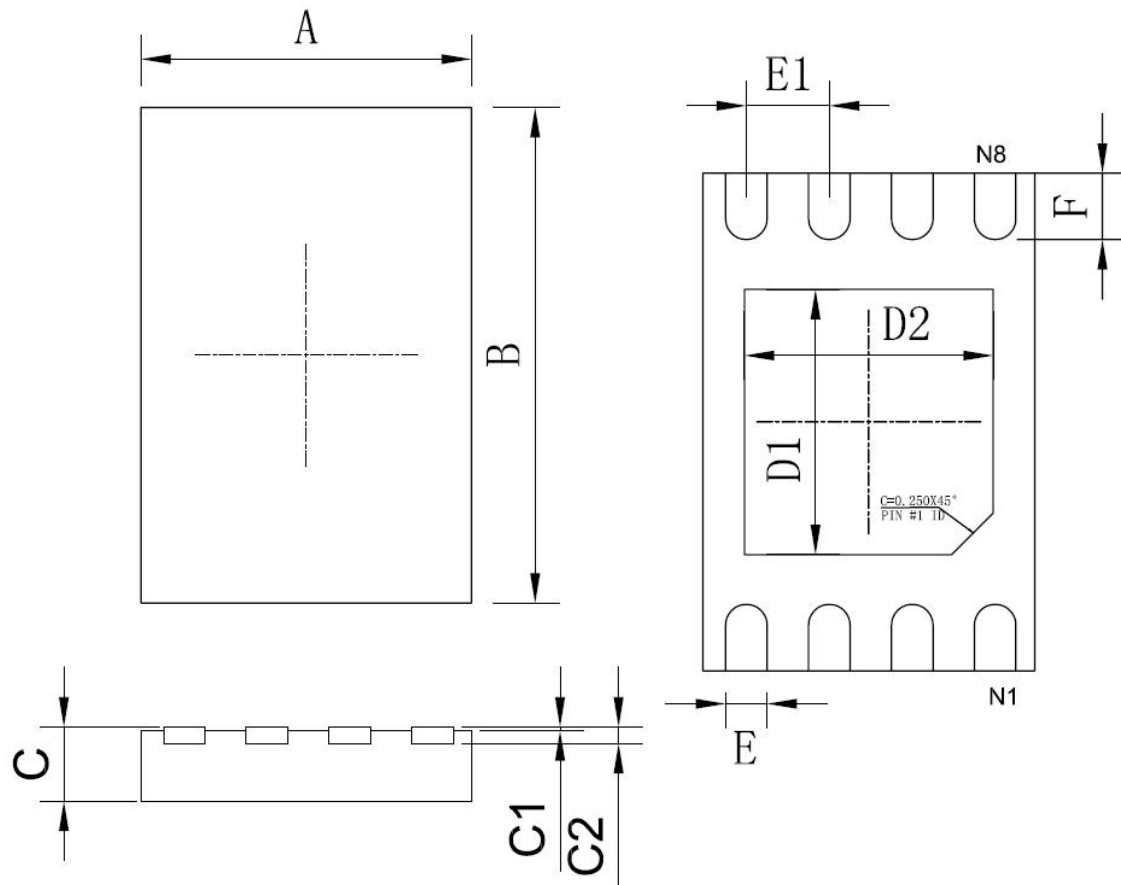


5 Package size

Package type	Package size
DFN8 2X3	2.0mmx 3.0mm (79milX118mil)
SOP8-B	5.28mmX7.90mm (208milX311mil)

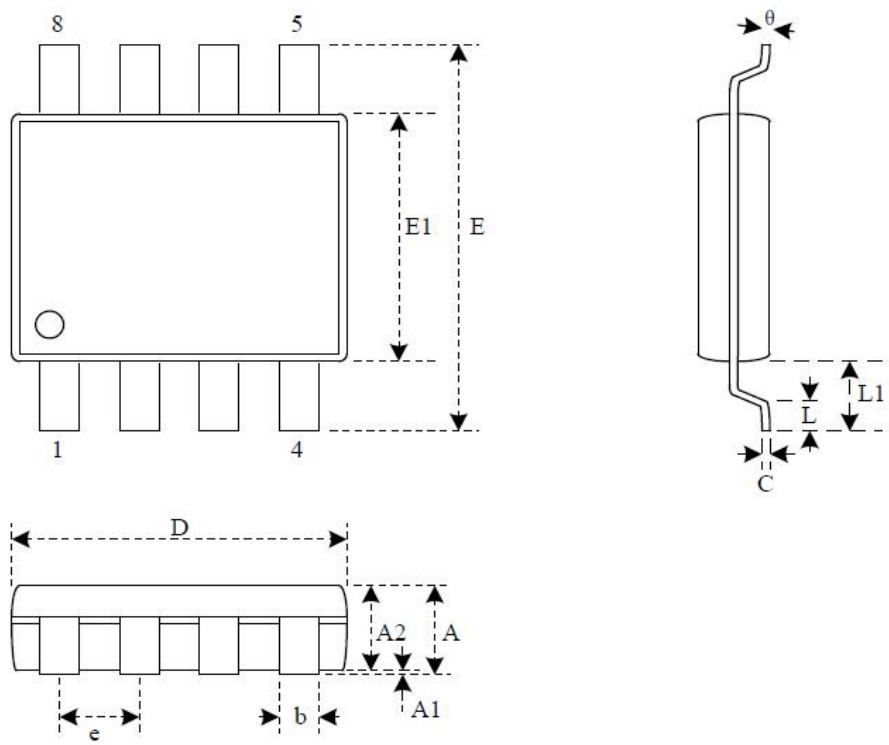
DFN8 2X3

Unit :mm



DIMENSION LABEL 尺寸 标注	MIN (mm) 最小 (mm)	MAX (mm) 最大 (mm)	DIMENSION LABEL 尺寸 标注	MIN (mm) 最小 (mm)	MAX (mm) 最大 (mm)
A	2.0±0.1		D1	1.60TYP	
B	3.0±0.1		D2	1.50TYP	
C	0.70	0.80	E	0.250TYP	
C1	0~0.050		E1	0.500TYP	
C2	0.203TYP		F	0.400TYP	

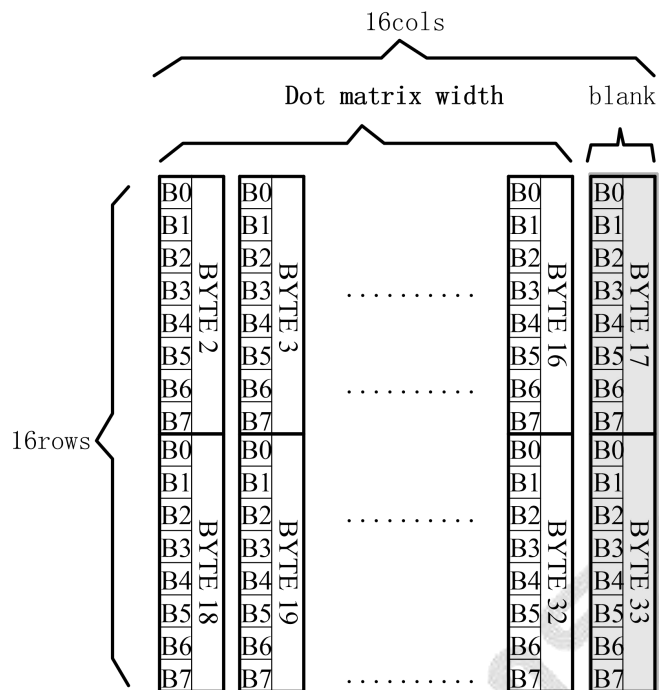
SOP8 -B



Symbol		A	A1	A2	b	C	D	E	E1		L	L1	
Unit													
Mm	Min.	-	0.05	1.70	0.36	0.19	5.13	7.70	5.18		0.50	1.21	0
	Norm.	-	0.15	1.80	0.41	0.20	5.23	7.90	5.28	1.27	0.65	1.31	5
	Max.	2.16	0.25	1.91	0.51	0.25	5.33	8.10	5.38		0.80	1.41	8
inch	Min.	-	0.002	0.067	0.014	0.007	0.202	0.303	0.204		0.020	0.048	0
	Norm.	-	0.006	0.071	0.016	0.008	0.206	0.311	0.208	0.050	0.026	0.052	5
	Max.	0.085	0.010	0.075	0.020	0.010	0.210	0.319	0.212		0.031	0.056	8

■ Storage structure

The dots matrix storage width of proportionally adjusted font uses BYTE as its unit. Different font.width will reveal corresponding blanks. With the font's actual width data stored in BYTE0~BYTE 1, it can be used as reference for the position of the next word.



For Example: ASCII Rounded font “B”

0-33 BYTE: 00 0C 00 00 00 00 00 00 7F 80 7F C0 60 C0 60 C0 60 C0 7F 80 7F C0 60 E0 60 60 60 60 7F C0 7F 80 00 00

In BYTE0~BYTE1: “00 0C” is width data, 12 bit width, 4 blank bits is reserved. The typeset of the next word may shift forward considering the blank bits.

In BYTE2~BYTE33: “00 00 00 00 00 00 7F 80 7F C0 60 C0 60 C0 60 C0 7F 80 7F C0 60 E0 60 60 60 7F C0 7F 80 00 00” is dot matrix data.

7 Dot data validation (customer reference)

Customers will chip "A" to call up the data with the following comparison. If so, it indicates SPI drive to work; if not, please re-write drivers.

Set row: Y (set vertical horizontal) dot size 8X16 Letter "A"

Lattice data: 00 E0 9C 82 9C E0 00 00 0F 00 00 00 00 00 0F 00





—Creating civilized intelligence

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